

DISCRETE LOGIC SERIAL MACROPAD

8-key keypad that emits a real 8-N-1 serial frame per key event, built only from 74HC glue logic and a diode ROM. No MCU, no FPGA, no firmware, no programmable part of any kind.

PRINCIPLE OF OPERATION

A 74HC161 sweeps a 3-bit address across the 8 keys at the bit clock, so every key is sampled 1/8 of the baud rate (1200 Hz at 9600 baud). 74HC151 #1 reads the LIVE debounced key at that address; 74HC151 #2 reads the STORED state of the same address out of a 74HC259 latch. A 74HC86 XORs them. A difference sets the BUSY flip-flop on the FALLING clock edge, which freezes the scan counter through its synchronous count-enables (CEP/CET) – the clock itself is never gated, so no runt pulses are possible. While BUSY, the frozen address selects one row of a 74HC238 + diode ROM, which presents 7 ASCII bits to a pair of 74HC165s. The shift registers emit start / b0..b6 / make-break / stop over the next 10 bit times. A second 74HC161 + 74HC138 count the frame, write the new key state into the 74HC259 at state 10, and release BUSY one bit later. Scanning resumes; N-key rollover falls out for free because each pending change is serviced on a later sweep.

OUTPUT JP4 selects 3.3V TTL UART or RS-232 (MAX3232 -> RJ45).
BAUD JP2 taps the 74HC4060 divider; see sheet 2 for the table.
ASCII MAP Set by which diodes are fitted; see sheet 6.



PWR_FLAGS: tell ERC that +3V3 / GND are driven.

Power, Clock, Reset

File: 01_power_clock.kicad_sch

Keys and Debounce

File: 02_keys_debounce.kicad_sch

Scanner and Memory

File: 03_scanner.kicad_sch

Transmit Sequencer

File: 04_sequencer.kicad_sch

ASCII Diode ROM

File: 05_ascii_rom.kicad_sch

UART Shift Register

File: 06_uart_tx.kicad_sch

Output Stage

File: 07_output.kicad_sch

Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

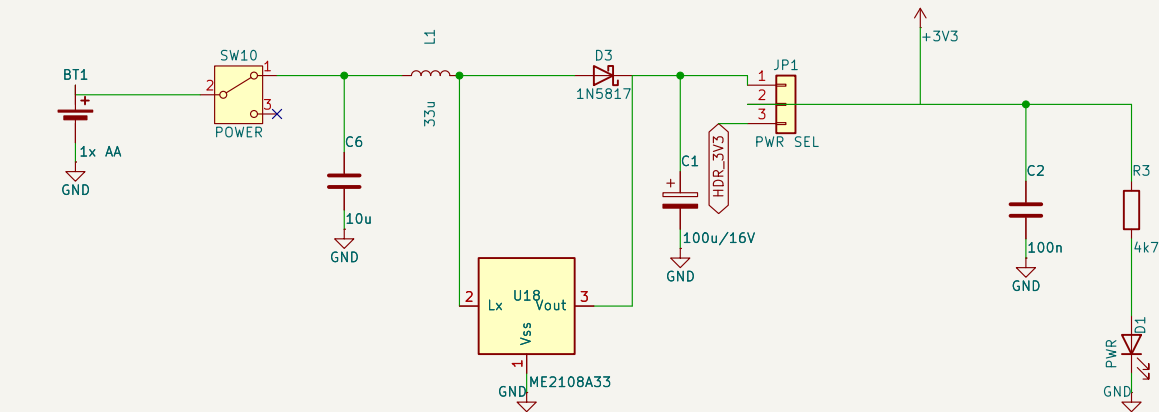
Sheet: /
File: logic-macropad.kicad_sch

Title: Logic Macropad – Root / block diagram

Size: A3	Date: 2026-09-04	Rev: A
KiCad E.D.A. 10.0.1		Id: 1/8

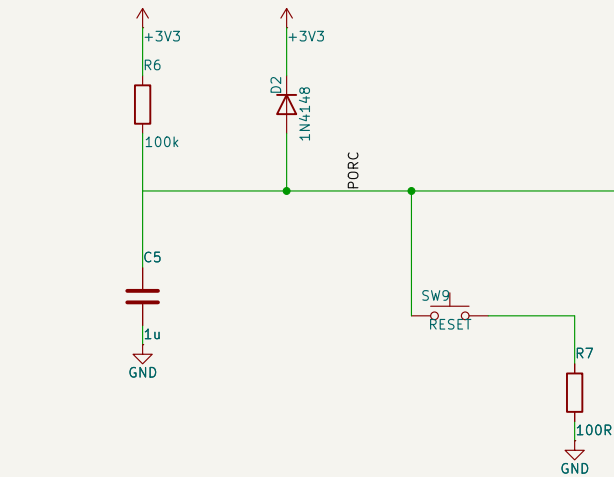
POWER INPUT

JP1 1–2 = powered from the AA cell through U18 (default)
2–3 = powered from the TTL header, J3 pin 4



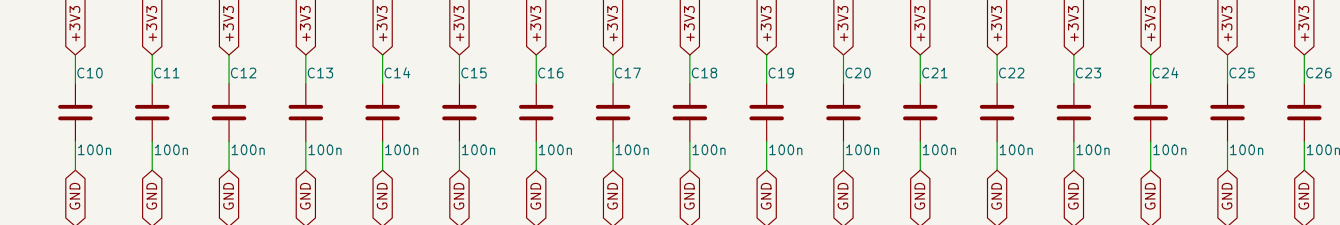
ME2108A33 needs exactly three parts around it:
L1 33 uH (DCR < 0.5 ohm), D3 1N5817, C1 as the output cap.
D3 sees 3.3 V reverse against its 20 V rating and ~100 mA of PFM peak against 1 A. The 20 V part has a LOWER forward drop than the 40 V 1N5819, and Vf comes straight off the output. Starts at 0.9 V and holds down to 0.45 V, so the cell is used to the end. 50 uA while switching, 9 uA once in regulation. SW10 breaks the battery lead outright – off is really off.

POWER-ON RESET

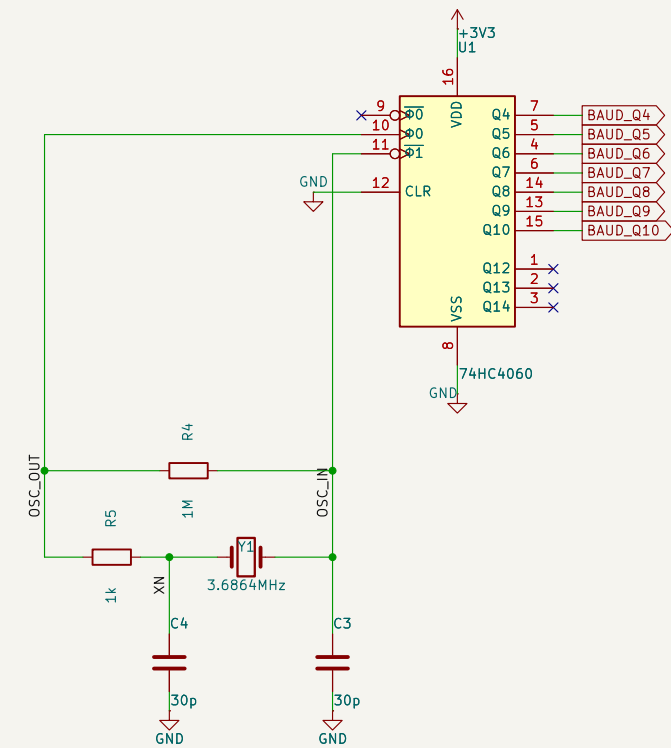


R6/C5 = 100 ms; the node crosses the 74HC14 threshold after ~87 ms, holding the 74HC259, the scan counter and the BUSY flip-flop cleared so no character escapes at switch-on. D2 dumps C5 into the falling rail at power-down, so a quick power cycle still starts from a discharged capacitor.

DECOUPLING – one per IC, at the VCC pin

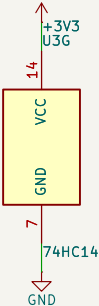
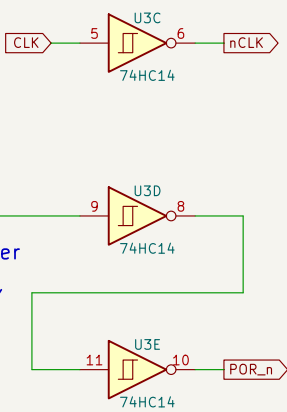


MASTER CLOCK / BAUD GENERATOR

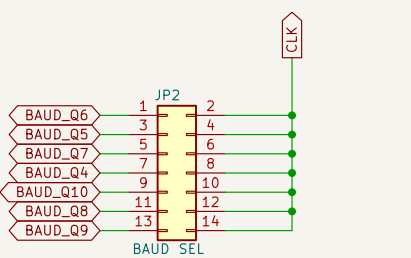


C3/C4 must be COG/NP0 – 2 x 30p gives CL = 20 pF

U3 74HC14 – units C..F stacked in pin order



U3A/U3B debounce KEY6 and KEY7 on sheet 3; U3C–E are above.
U3F debounces KEY5 on sheet 3, beside its own RC cell.



BAUD SELECT (JP2) – fit ONE shunt across a column

The columns are NOT in baud order. The 4060 does not carry its taps in pin order, and the columns follow the PACKAGE so that the seven wires to U1 do not cross. Read this table, not the position:
col 1 = Q6 (/64) col 2 = Q5 (/32) col 3 = Q7 (/128)
col 4 = Q4 (/16) col 5 = Q10 (/1024) col 6 = Q8 (/256)
col 7 = Q9 (/512)

Y1 = 3.6864 MHz (FITTED) col2=115200 DEFAULT col1=57600
col4=230400 col3=28800 col6=14400
Y1 = 4.9152 MHz (ALT) col7=9600 col6=19200 col3=38400

RouterBOOT on an RB4011 talks 115200 and that rate is not settable, so a 9600-only board cannot read the boot messages. 115200 and 9600 differ by 12, not a power of two, so no one crystal reaches both – swapping Y1 is the only way across.

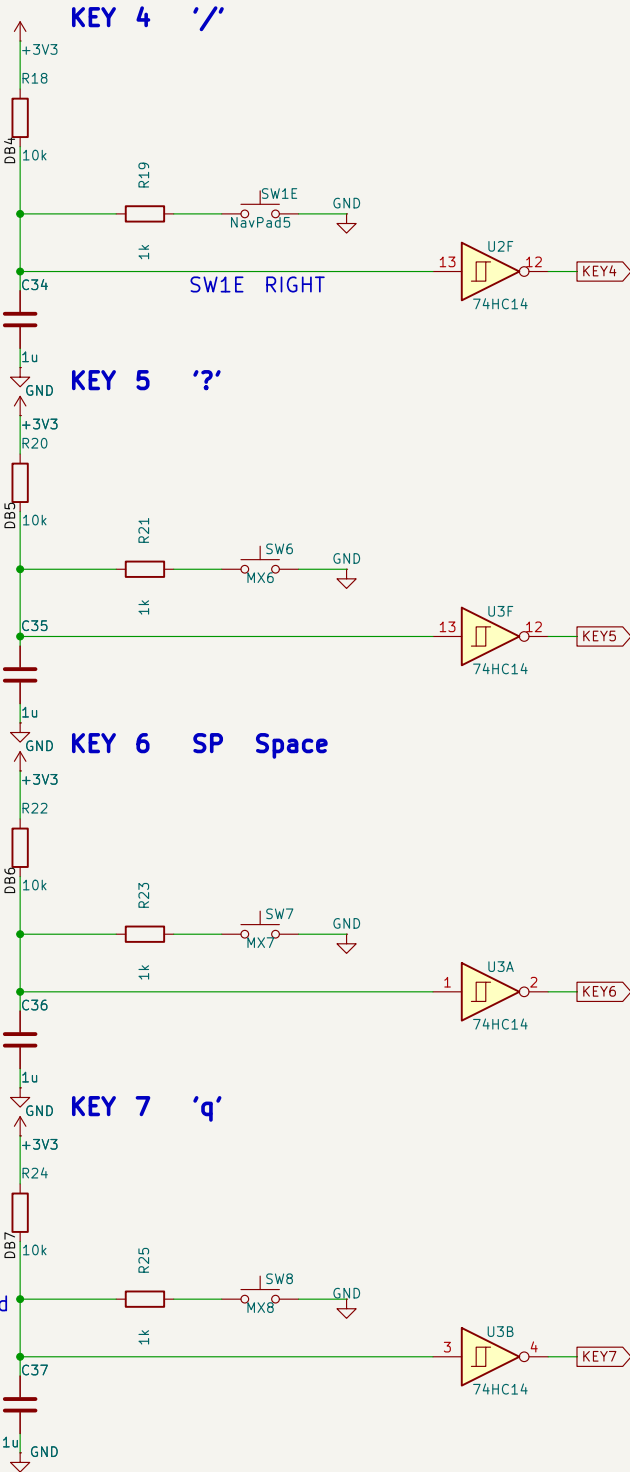
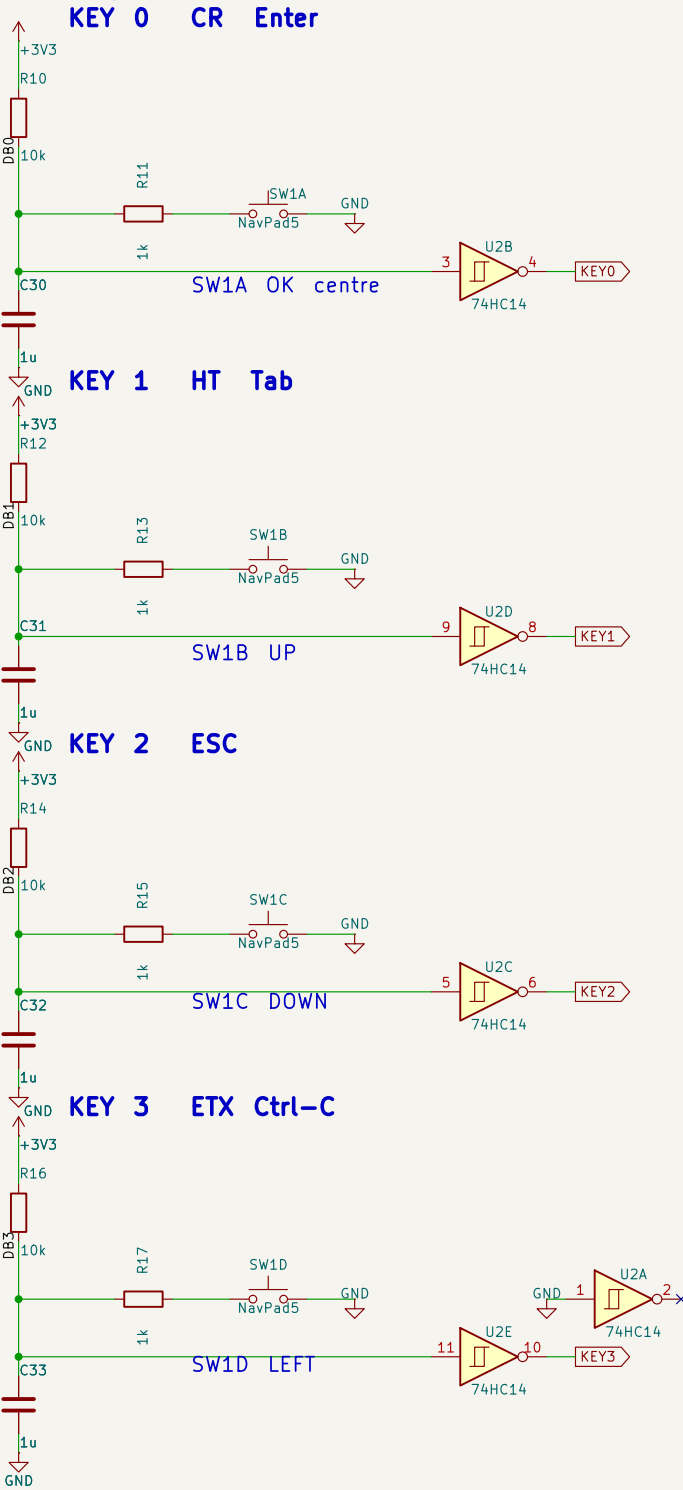
1.8432 MHz was the original choice but has no ordinary SMD crystal: HC-49/US SMD starts at about 3.2 MHz. Doubling it to 3.6864 MHz just moves the default tap from Q4 to Q5.

Selectable TTL UART or RS-232 output 8-key macropad, 74HC logic only, no MCU/FPGA		
Sheet: /Power, Clock, Reset/ File: 01_power_clock.kicad_sch		
Title: Logic Macropad – Power, Clock, Reset		
Size: A3	Date: 2026-09-04	Rev: A
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KEY INPUT – eight independent debounced switches

Each cell is drawn in full: 10k pull-up, 1k series, 1uF to ground, into a 74HC14 Schmitt inverter. Press discharges through the 1k (tau = 1 ms); release recharges through the 10k (tau = 10 ms). The asymmetry is deliberate – the slow edge swallows contact bounce, and guarantees two changes on one key are milliseconds apart, far longer than the 11 bit times the transmitter needs. Without the Schmitt a slow ramp into a CMOS input chatters, so the 74HC14 is not optional. Use 10V or 16V 1uF parts: DC bias costs a 6.3V 0402 much of its 1uF.

KEY0..KEY4 are the five contacts of one SJMS 5-way cluster (SW1 units A..E, one footprint); KEY5..KEY7 are Cherry MX.



Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

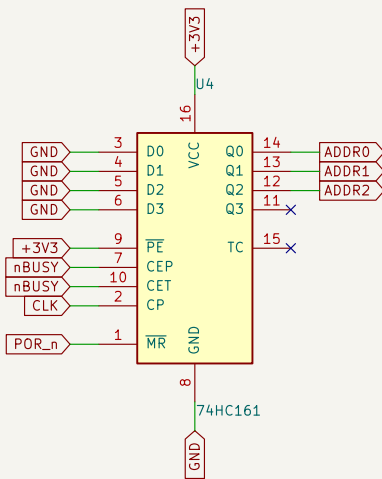
Sheet: /Keys and Debounce/
File: 02_keys_debounce.kicad_sch

Title: Logic Macropad – Keys and Debounce

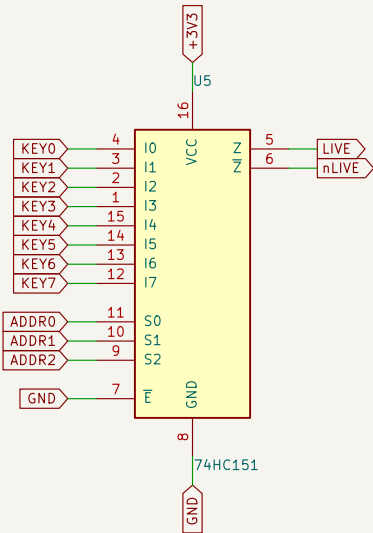
Size: A3 Date: 2026-09-04
KiCad E.D.A. 10.0.1

Rev: A
Id: 3/8

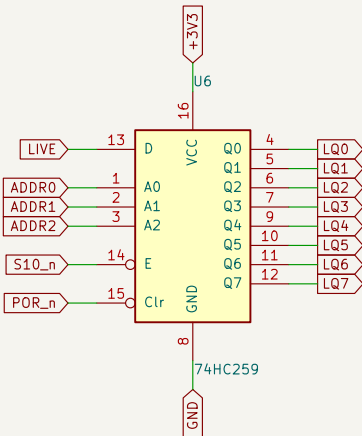
SCANNER, STATE MEMORY AND CHANGE DETECTION



Counts on CLK rising edges. CEP/CET = nBUSY: the counter is stopped by a synchronous ENABLE, never by gating the clock. Q3/TC unused.

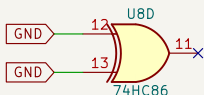
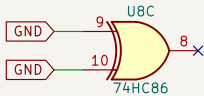
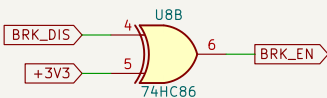
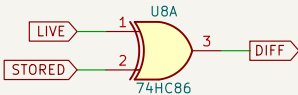


LIVE = physical state of the addressed key.



Written once per frame, during sequencer state 10, i.e. after the stop bit has been shifted out. POR_n clears all 8 bits to 0 at power-up so the stored state matches "no key pressed".

DIFF = LIVE xor STORED



U7 is the part the original sketch was missing. A 74HC259 has no read-back port: its 8 latches are only visible as 8 parallel outputs, so a second multiplexer is required to look at the one bit the scanner is currently addressing.

U8B: an XOR with one input tied high is an inverter. BRK_EN = NOT BRK_DIS, used to gate frame bit 7 (sheet 5).

Spare XOR gates parked with both inputs at GND.

Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

Sheet: /Scanner and Memory/
File: 03_scanner.kicad_sch

Title: Logic Macropad – Scanner and Memory

Size: A3 Date: 2026-09-04
KiCad E.D.A. 10.0.1

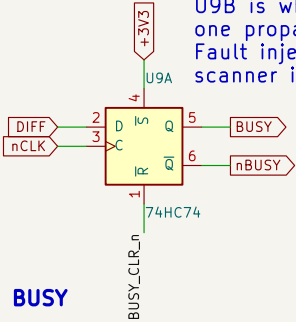
Rev: A
Id: 4/8

TRANSMIT SEQUENCER AND CONTROL

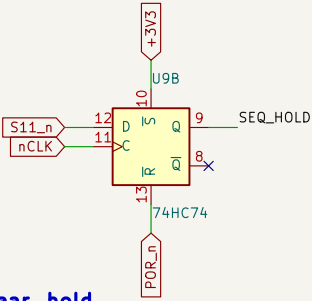
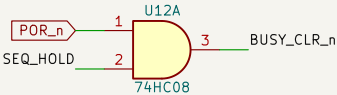
All of this runs on nCLK (the falling edge of the bit clock).

f0 BUSY set from DIFF. Scan counter freezes on the changed key.
~PL of the 74HC165s releases, so the START bit is already on TX.
f1..f9 nine shifts: b0..b6, make/break, stop. Ten bit times total.
f10 sequencer state 10 -> ~S10 pulses the 74HC259 write enable, so the new key state is stored. DIFF collapses to 0.
f11 With DIFF now 0, U9A clocks BUSY low from its own D input.
The scanner resumes on the next CLK edge. Total halt: 11 bit times - the frame plus the one cell that writes the latch.

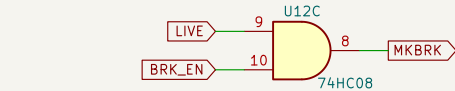
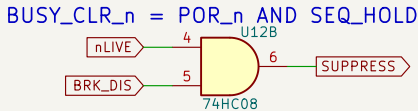
STATES 11-12 ARE A LOCK-UP ESCAPE, NOT PART OF THE NORMAL PATH. They only run if DIFF somehow fails to collapse at state 10, which would otherwise leave BUSY stuck and the scanner frozen for good. Then ~S11 goes low, U9B captures it on the next edge and drives SEQ_HOLD low, asynchronously clearing BUSY at f12. Clearing BUSY clears the sequencer, so ~S11 goes away again - registering it in U9B is what holds the clear asserted for a full bit time instead of one propagation delay, and removes the self-clearing race. Fault injection in tools/simulate.py jams DIFF high and confirms the scanner is freed after 12 bit times.



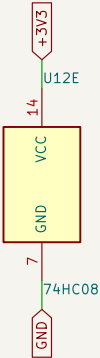
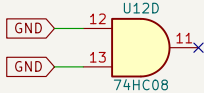
U9A = BUSY



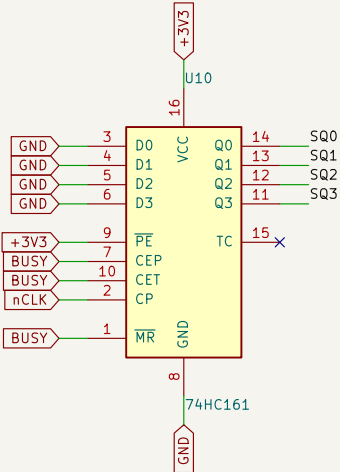
U9B = clear hold



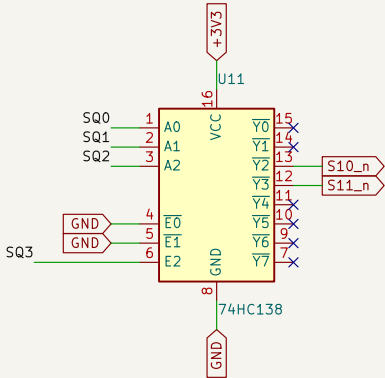
SUPPRESS = nLIVE AND BRK_DIS
(blank the frame on key RELEASE when break codes are disabled)



MKBRK = LIVE AND BRK_EN -> frame bit 7.
With JP3 open (press only) bit 7 is forced to 0, so the console receives clean 7-bit ASCII rather than 0x80|code. With JP3 fitted, bit 7 is 1 on press and 0 on release.

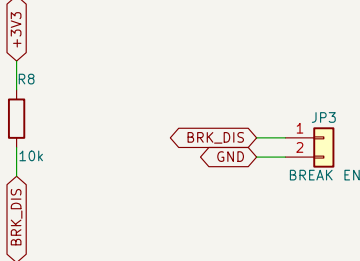


~MR = BUSY: the sequencer sits at state 0 whenever the transmitter is idle.



E2 = SQ3, so Y0..Y7 decode states 8..15. Only ~Y2 (=state 10) and ~Y3 (=state 11) are used; no inverter is needed because ~Y2 drives the active-low write enable of the 74HC259 directly.

BREAK CODE OPTION



JP3 OPEN (default): press only. Bit 7 always 0, release frames are blanked. This is what a RouterOS console wants - 0x80|ASCII is garbage to it.
JP3 FITTED: full make/break. Bit 7 = 1 on press, 0 on release.

Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

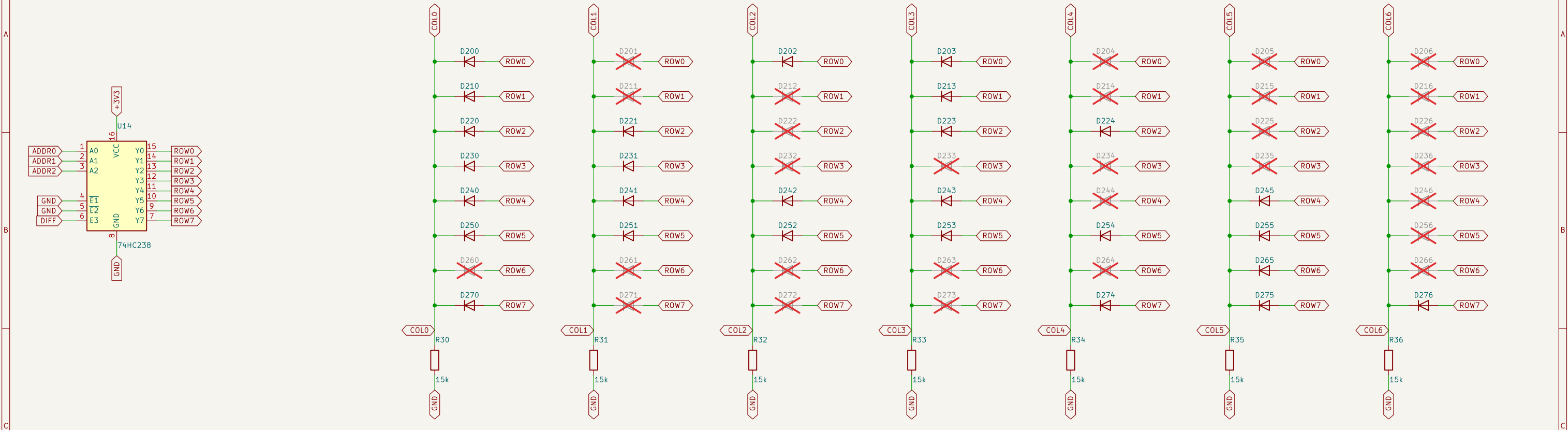
Sheet: /Transmit Sequencer/
File: 04_sequencer.kicad_sch

Title: Logic Macropad - Transmit Sequencer

Size: A3 Date: 2026-09-04
KiCad E.D.A. 10.0.1

Rev: A
Id: 5/8

ASCII DIODE ROM – 8 rows x 7 bits



'238, not '138: the outputs must go HIGH to push current through the diodes into the pull-downs. A 74HC138 has active-LOW outputs and would need pull-ups and the opposite diode orientation.

E2 = DIFF, so the ROM is powered only while a key change is pending. DIFF asserts within ~60 ns of the scan step and the columns settle in about 1.4 us, so the data is valid long before the 74HC165 load ends half a bit later (52 us at 9600 baud, 4.3 us at 115200). Driving E2 from BUSY instead would be a bug: BUSY and the parallel load change on the same edge, and the shift registers would latch 0.

Cathode -> column, anode -> row. Unselected rows sit at 0 V, so their diodes are reverse biased: no sneak paths. A selected column sits one 1N4148 drop below the 3.3 V row: about 2.76 V (0.54 V at 184 uA into 15k) against the 2.31 V VIH of the 74HC165 – 450 mV of margin. 10k works as well (2.74 V); BAT54 Schottkys add ~0.3 V. See tools/check_levels.py.

DEFAULT MAP – fit the diodes marked "1" (27 of 56):

key	code	char	b6	b5	b4	b3	b2	b1	b0
K1	0x0D	CR Enter	0	0	0	1	1	0	1
K2	0x09	HT Tab	0	0	0	1	0	0	1
K3	0x1B	ESC	0	0	1	1	0	1	1
K4	0x03	ETX Ctrl-C	0	0	0	0	0	1	1
K5	0x2F	'/'	0	1	0	1	1	1	1
K6	0x3F	'?'	0	1	1	1	1	1	1
K7	0x20	SP Space	0	1	0	0	0	0	0
K8	0x71	'q'	1	1	1	0	0	0	1

Changing the map = moving diodes. That is the whole point.

Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

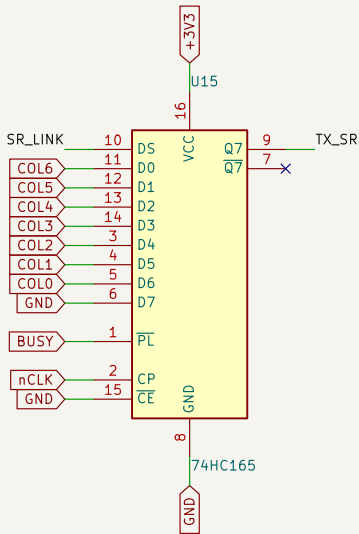
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File: 05_ascii_rom.kicad_sch
Title: Logic Macropad – ASCII Diode ROM

Size: A3	Date: 2026-09-04	Rev: A
KiCad E.D.A. 10.0.1		Id: 6/8

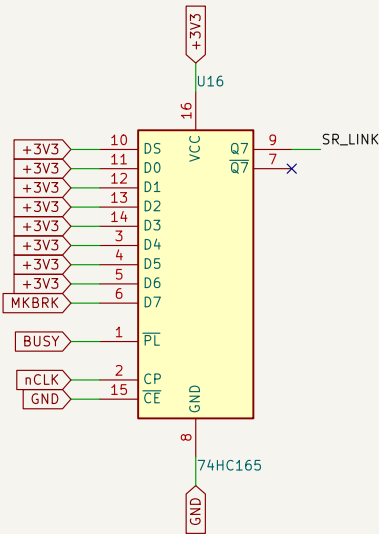
UART TRANSMITTER – 10-bit frame from two 74HC165

A 74HC165 shifts towards Q7 and presents input D7 first, so the parallel inputs are loaded in reverse frame order.
Frame: 0 (start) . b0 b1 b2 b3 b4 b5 b6 . make/break . 1 (stop)
That is a genuine 8–N–1 byte whose bit 7 carries press/release.

~PL is tied straight to BUSY: while idle the registers are held in parallel-load, so the start bit is standing ready on Q7 and appears the instant BUSY rises. ~CE is tied low – the asynchronous load overrides the clock, so no shift-inhibit gating is needed.

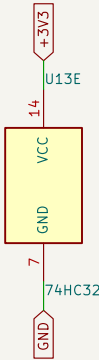
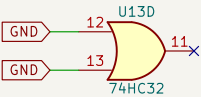
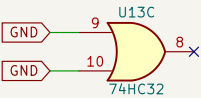
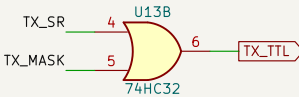
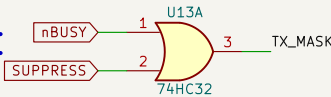


U15: start bit + ASCII b0..b6
D7=start(0) D6=b0 D5=b1 D4=b2
D3=b3 D2=b4 D1=b5 D0=b6



U16: make/break + stop + idle ones
D7=make/break D6=stop(1)
D5..D0 = 1 so the line idles high after the frame.

TX OUTPUT GATE



TX_TTL = TX_SR OR nBUSY OR SUPPRESS.
nBUSY parks the line high between frames; SUPPRESS blanks a whole release frame when JP3 is open.

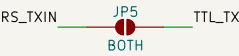
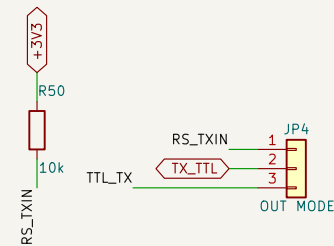
Selectable TTL UART or RS-232 output
8-key macropad, 74HC logic only, no MCU/FPGA

Sheet: /UART Shift Register/
File: 06_uart_tx.kicad_sch
Title: Logic Macropad – UART Shift Register

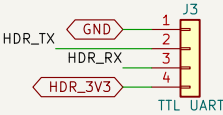
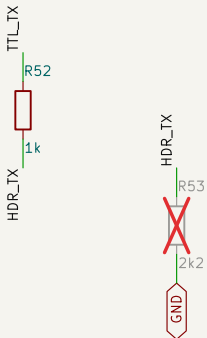
Size: A3 Date: 2026-09-04 Rev: A
KiCad E.D.A. 10.0.1 Id: 7/8

OUTPUT STAGE – 3.3 V TTL UART or RS–232

JP4 is the mode selector. Its centre pin is the live TX from the logic; each outer pin feeds one output driver and is parked high by a 10k so the unselected output sits at its idle level instead of floating.
JP4 1–2 : RS–232 on the RJ45. TTL header idles high.
JP4 2–3 : 3.3 V TTL on J3. RS–232 driver parked at MARK.
JP5 (solder bridge) shorts pin 1 to pin 3 -> both outputs live.



TTL UART HEADER



J3 1 GND
2 TXD (macropad -> host, idles high)
3 RXD (console output monitor, via the MAX3232 receiver)
4 +3V3 (bus power in/out, routed through JP1)

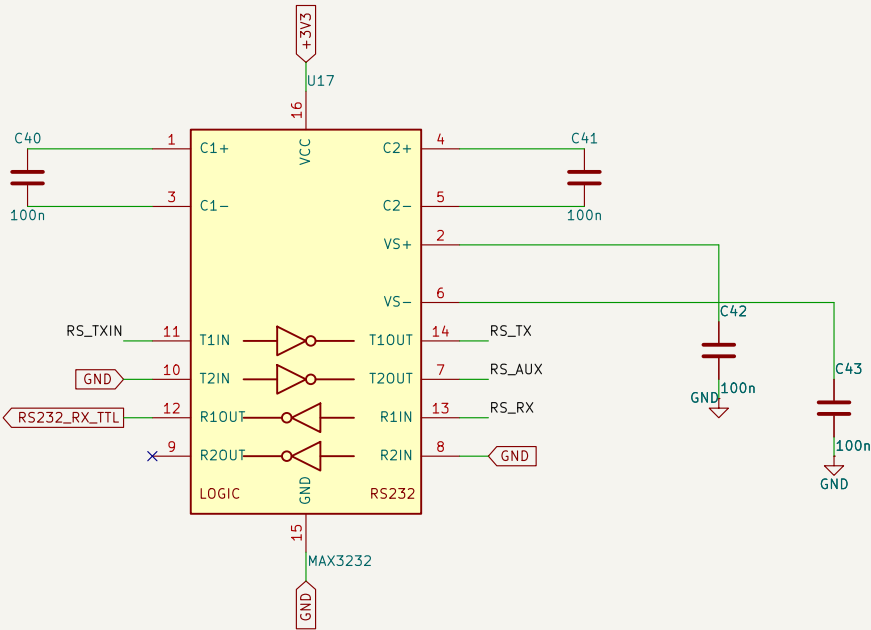
R53 is NOT fitted, and at 3.3 V it should stay that way.
 $R52/R53$ divide by $2k2/(1k+2k2) = 0.69$, which was sized when this rail was 5 V: $5 \times 0.69 = 3.4$ V, the number this note used to quote. The rail is 3.3 V now, so fitting R53 would deliver 2.3 V – below the 2.4 V a 3 V CMOS input wants to see, and no help to a 5 V host either. Drive it straight through R52.

Cisco–style console pinout, which is what an RB4011 expects:
1 RTS(out) 2 DTR(out) 3 TXD(out) 4 GND 5 GND
6 RXD(in) 7 DSR(in) 8 CTS(in)
Pins 1 and 2 are OUTPUTS of the router and are deliberately left unconnected – driving them would fight the router.

JP6/JP7 both on 2–3 : straight patch cable to the router (default).
JP6/JP7 both on 1–2 : rollover console cable.
Never select the same RJ45 pin on both jumpers.

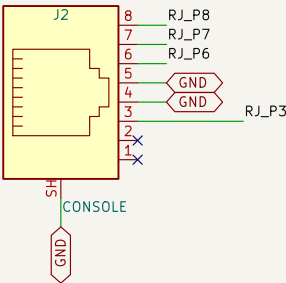
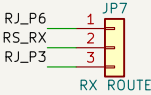
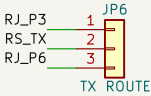
JP8/JP9 are open by default; bridge them only if a device refuses to talk without DSR/CTS asserted. RouterOS does not need them.

RS–232 LEVEL SHIFTER



MAX3232 / MAX232A: 100 nF charge–pump caps.
A genuine MAX232 (no suffix) needs 1 uF – the original spec called for 0.1 uF with a MAX232, which will not pump.

CONSOLE PORT



Selectable TTL UART or RS–232 output
8–key macropad, 74HC logic only, no MCU/FPGA

Sheet: /Output Stage/
File: 07_output.kicad_sch

Title: Logic Macropad – Output Stage

Size: A3 Date: 2026–09–04
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